

Variational Autoencoders

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1 Introduction

About one month ago, I re-read Chapter 30 of *Foundations of Computer Vision* Torralba et al. [2024] which covers the concept of representation learning. In this chapter, Torralba presents a learning problem that feels ever so familiar when viewed from a certain perspective¹. Formally, that is to take an input $x \in X$ and compose it into an abstract representation $z \in Z$.

Think of an orange cube. What we picture in our mind when thinking on this cube, is an abstract representation of it. That is, there isn't a literal cube present, that we can observe through our senses. The cube exists in our mind, and due to our familiarity with something as simple as a cube, our mental image is incredibly rich.

Our representation is so rich that if you think again on your original orange cube, you could very easily color it blue, green, or red and maybe even place a ball on top. Our mind is incredibly well suited at composing basic concepts and forms to construct more complex representations. After all, this compositional process is the foundation of how we form knowledge. The cube exercise is especially easy to appreciate because it contains some of the most fundamental abstract concepts in colors, shapes, and perspective.

Here, we can return to our originally defined learning problem with a sense of familiarity. The learned representation, z , what we are after, is an abstraction of the data, x , much like our internal representation of a cube is to that of a cube, which could be immediately perceived.

The representation learning problem presents an interesting opportunity to study how we construct the abstract representation z and how it can be composed for later use. Namely, by studying the representation of an abstraction we can better interpret how downstream judgements are made.

All methods of forming representations make use of pretext tasks that guide the formation of the structure of the abstract space. Through this, we are able to control for the information that is transmitted or ignored. These tasks are either compressive or predictive. In predictive tasks, we use labels to form the representation that is most relevant to the given task. Namely, we impose a preexisting structure over the representation which is most useful for a completing a task. Then, compressive tasks encourage the model to form its own representations by fitting the data through a bottleneck. More simply, we ask the model to compress the data into a limited space, keeping only the information needed to reconstruct it. We can think of compressive tasks as an approach that embodies *Occam's Razor* Torralba et al. [2024].

The primary focus of my project is to explore compressive tasks, while briefly inspecting predictive tasks to compare how they differ in their representational approach. Specifically, I focus this exploration on the Variational Autoencoder (VAE). The VAE's goal is to take an input x , encode it into z , and then attempt to reconstruct x based on our abstract representation z . Through this process, the model learns a latent representation that preserves what is essential for reconstruction, while being forced into a simple and compact form. It captures the core structure of the data, much like a mental representation captures the essence of an object rather than every detail.

I explore the VAE through its original paper and supporting work to establish the technical foundation of this project. I then conduct experiments on the WikiArt dataset Tan et al. [2019] to study how different

¹Throughout this report I include some thought experiment blurbs that I found useful in forming my intuitive understanding of the topics presented

VAE variants construct their latent spaces. An art dataset is particularly well suited for this purpose, as it presents a rich and inherently abstract domain. By examining how a model represents paintings, we gain insight into how it interprets visual structure, which in turn allows us to ask why it makes certain judgments.

Think of a VAE as a painter. Suppose this painter seeks to study a landscape and paint it. When studying the landscape, the artist decomposes the painting into its simplest forms, such as shapes, color, and lighting. This allows the painter to paint it not from its explicit form, but from an abstract representation that exists in their mind.

A landscape is harder to abstract than an orange cube, but we can still try to do so. Think on a vista you have recently seen, and attempt to picture it in your mind. Due to the complexity and richness of the vista, it may not be as easy to picture. Certainly not as easy as a cube. The difficulty of this exercise is what interests me most when studying it through a VAE.

Some of these experiments include performing latent space² probing and classification tests to study the quality and organization of the latent embeddings. Beyond just reviewing VAEs, I sought to answer two questions that arose during my exploration:

- (1) How well does a VAE, in all its forms, organize and compress data into an abstract space?
- (2) How useful is this abstract space in downstream tasks, such as classification?

I must make clear that this project does not propose a new model; rather, it synthesizes existing theory and uses empirical analysis as a lens for interpretation. Fundamentally, I sought to explore the principles underlying generative modeling and how one of the most foundational models in this space approaches the representation learning problem.

2 Background

This section does not provide a comprehensive review of the works supporting VAEs and representation learning, it highlights the material that best helps in constructing the theoretical and conceptual framework of this project.

Before introducing the probabilistic formulation of VAEs, it is useful to frame the representation learning problem through an information theoretic lens. Doing so allows us to formalize what it means to *compress* data while preserving what is relevant. This perspective will later reveal that the VAE objective is not just a probabilistic construction, but a specific instance of a broader information theoretic principle. I start by first highlighting the works in Tishby et al. [2000] that established the original formulation of the information bottleneck method. Given an input data source $x \in X$ and labels $y \in Y$, where our dataset is $D = \{(x_i, y_i)\}_{i=1}^N$, we seek to minimize Eqn. 1.

$$L(p(\tilde{x} | x)) = I(\tilde{X}; X) - \beta I(\tilde{X}; Y) \quad (1)$$

where $I(\cdot; \cdot)$ denotes mutual information, β is a Lagrange multiplier, and $\tilde{X}$ is our compressed representation of X . As later elaborated on in Tishby and Zaslavsky [2015], we seek a maximally compressed form of $\tilde{X}$ that preserves the most amount of information about Y , while fitting X through a bottleneck. The labels Y impose a structure on the representation of X in $\tilde{X}$, allowing $\tilde{X}$ to be interpreted as a minimal sufficient statistic of X with respect to Y .

The information bottleneck can be understood through the analogy of studying for an exam. The course material is the input X , while practice problems and answer keys play the role of labels Y . Effective studying does not require memorizing every detail of the material; instead, one compresses the content into the concepts most relevant for answering questions. The learned understanding is a compressed representation of X that preserves what is necessary to predict Y .

The minimization in Eqn. 1 can equivalently be written as a constrained maximization problem,

$$\max_{p(\tilde{x}|x)} I(\tilde{X}; Y) \quad \text{s.t.} \quad I(\tilde{X}; X) \leq I_c, \quad (2)$$

²I often interchange latent and abstract space, but they describe exactly the same thing.

for some information budget I_c . Introducing a Lagrange multiplier recovers the same objective up to a sign change, i.e.,

$$\max I(\tilde{X}; Y) - \beta I(\tilde{X}; X), \quad (3)$$

which is equivalent to Eqn. 1. The classical formulation optimizes directly over conditional distributions $p(\tilde{x}|x)$, and thus characterizes the optimal tradeoff between compression and prediction at the level of information theory. However, computing the mutual informations requires access to the true joint distribution $p(x, y)$ and optimizing over arbitrary conditional distributions. In deep learning settings, this becomes intractable thus, a parametric and tractable relaxation is required.

This motivates an approach leveraging variational approximation, where we can conveniently look towards Alemi et al., 2016, which does exactly that. Rather than optimizing over arbitrary $p(\tilde{x}|x)$, we parameterize a stochastic encoder $p_\theta(z|x)$ with neural network parameters θ , thereby restricting the search space to a tractable family of encoders. We then seek to find a latent encoding Z that is both predictive of our labels Y and maximally compressed with respect to X . Formally, we tackle the constrained maximization problem defined in Eqn. 4:

$$\max_{\theta} I_\theta(Z; Y) \quad \text{s.t.} \quad I_\theta(Z; X) \leq I_c \quad (4)$$

where I_c is the information constraint that encourages compression in the latent embedding. Introducing a Lagrange multiplier yields

$$R_{IB}(\theta) = I_\theta(Z; Y) - \beta I_\theta(Z; X), \quad (5)$$

which is the variational information bottleneck objective. Note that this maximization form is equivalent to the earlier minimization in Eqn. 1 up to a sign change in the objective and the substitution $\tilde{X} \rightarrow Z$.

What remains to be understood is how the information-theoretic objective connects to practical generative models trained with neural networks. The Variational Autoencoder (VAE) provides precisely this bridge, allowing us to examine the central paper to this project in its original formulation Kingma and Welling [2013]. The VAE can be viewed as a probabilistic realization of the same compression–prediction tradeoff described above, but instantiated through likelihood maximization and variational inference. Here, we seek to minimize the variational lower bound on the marginal likelihood of a datapoint (x_i) , which can be seen in the following objective:

$$L(\theta, \phi; x_i) = -D_{KL}(q_\phi(z|x_i)||p_\theta(z)) + \mathbb{E}_{q_\phi(z|x_i)}[\log p_\theta(x_i|z)] \quad (6)$$

where D_{KL} is the Kullback–Leibler divergence between the prior $p_\theta(z)$ and the approximate posterior $q_\phi(z|x_i)$. This loss, known as the Evidence Lower Bound (ELBO), accomplishes two things. Given an input x_i , we train a neural network q_ϕ to estimate a latent encoding $z \sim q_\phi(z|x_i)$ that is both informative of the true prior distribution while being predictive enough to accurately reconstruct the original input $\hat{x}_i \sim p_\theta(x_i|z)$. This dual goal should feel familiar.

Earlier, from the information theoretic perspective, we sought a minimal sufficient statistic Z of X with respect to labels Y . In the autoencoding setting, the reconstruction target plays the role of Y , making the relevance term self-referential ($Y = X$). Alemi et al. [2016] makes this connection explicit by showing that the ELBO can be interpreted as a variational bound on an information bottleneck objective with $Y = X$. In particular, the reconstruction term can be interpreted as providing a lower bound on the mutual information between Z and X under suitable assumptions, while the KL term upper bounds $I(Z; X)$ through a divergence to the prior. In practice, these mutual information terms are intractable, requiring variational approximations which yields,

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta D_{KL}(q_\phi(z|x)||p_\theta(z)). \quad (7)$$

Thus, the VAE objective can be understood simultaneously as (i) a lower bound on log-likelihood and (ii) an instance of the information bottleneck principle where the prediction target is the data itself ($Y = X$). From this perspective, the KL term already plays the role of a compression penalty, directly analogous to the $I(Z; X)$ term in Eqn. 5. The ELBO therefore embodies the same compression–relevance tradeoff introduced in the original information bottleneck formulation. It is within this balance that we can study methods to control latent structure while preserving predictive fidelity.

Having made this connection explicit, the introduction of β -VAE [Higgins et al. [2017], Burgess et al. [2018]] appears not as a fundamentally new objective, but as a direct reinforcement of the compression parameter β that already appeared in Eqns. 1 and 5. By reweighting the KL term in the ELBO (Eqn. 6),

$$L(\theta, \phi; x) = -\beta D_{KL}(q_\phi(z|x)||p_\theta(z)) + \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] \quad (8)$$

the model explicitly controls the strength of compression in the latent representation. When $\beta = 1$, we recover the vanilla VAE objective. As we increase β , we encourage stronger compression, which often leads to more disentangled latent representations at the cost of reconstruction fidelity. From the information-theoretic perspective, this behavior is expected: increasing β shifts the model along the information plane toward lower $I(Z; X)$, sacrificing reconstruction fidelity in exchange for more constrained and often more structured representations.

Then, in Zhao et al. [2017], the authors highlight two shortcomings of the original VAE formulation. First, they show that minimizing the standard ELBO does not guarantee alignment between the aggregated posterior $q_\phi(z)$ and the prior $p_\theta(z)$, potentially leading to mismatched latent structure. Second, if the decoder $p_\theta(x|z)$ is sufficiently expressive, the latent variables may be ignored entirely. To address this, they introduce a modified objective:

$$L(\theta, \phi; x) = -(1 - \alpha) D_{KL}(q_\phi(z|x)||p_\theta(z)) + \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - (\alpha + \lambda - 1) D_{KL}(q_\phi(z)||p_\theta(z)) \quad (9)$$

and since the aggregated posterior $q_\phi(z)$ is intractable, they replace the final KL term with a Maximum Mean Discrepancy (MMD) distance Gretton et al. [2006], yielding the final InfoVAE objective:

$$L(\theta, \phi; x) = -(1 - \alpha) D_{KL}(q_\phi(z|x)||p_\theta(z)) + \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - (\alpha + \lambda - 1) D_{MMD}(q_\phi(z)||p_\theta(z)) \quad (10)$$

Having established the theoretical connection between compression, representation, and generative modeling, we now turn to an empirical investigation of these ideas. In particular, we study how different VAE variants organize their latent spaces and how this organization impacts both reconstruction and downstream classification. The following section outlines the experimental setup used to explore these questions.

3 Methods

As a reminder, I highlight the two central questions guiding the experimentation for this project.

- (1) How well does a VAE, in all its forms, organize and compress data into a latent space?
- (2) How useful is this latent space in downstream tasks such as classification?

To answer these questions, I first highlight our dataset and present why it is an interesting candidate of study. Then, I present the various architectures trained, including a description of the training approach. Finally, I present both quantitative and qualitative approaches to assist in answering our questions. There, I more clearly define the *quality* and *usefulness* of the latent space.

3.1 Dataset

I make use of a precompiled WikiArt dataset from Tan et al. [2019], consisting of over 40,000 paintings across 22 styles. Figure 1 showcases some examples from the dataset that span styles such as Impressionism, Expressionism, Abstract Expressionism, High Renaissance, Realism, and more. A natural question to pose regarding this dataset is the following: given a painting, can we correctly identify its artistic style? Formally, this defines the central downstream task considered in this project. Given an input image x , the goal is to predict its corresponding style label $y \in \{1, \dots, K\}$, where each class represents a distinct artistic style. This classification task serves as a proxy for evaluating whether the learned latent representation can be composed to capture meaningful stylistic structure.

This motivates both quantitative and qualitative analyses: classification experiments evaluate whether the learned structure is useful, while latent space analysis examines how stylistic *understanding* emerges within the representation.



Figure 1: Examples From the WikiArt Dataset Tan et al. [2019]

3.2 Modeling Approach

I consider two model families. The first is a standard convolutional VAE, which serves as a controlled and interpretable baseline. The second is a deeper variant, which we can refer to as *DeepVAE*, which incorporates residual connections to increase representational capacity and improve optimization stability He et al. [2016]. Across both architectures, I explore three objective formulations: the vanilla VAE, β -VAE, and InfoVAE objectives. Rather than exhaustively exploring all configurations at scale, I first sweep a range of hyperparameters on the base VAE architecture and then select representative configurations for deeper experimentation.

Objective	Parameter	Values Explored
Vanilla VAE	β	$\{1\}$
β -VAE	β	$\{5, 10, 25, 50\}$
InfoVAE	λ	$\{50, 75, 100, 250, 500\}$

Table 1: Hyperparameter configurations explored for VAE variants.

Based on validation behavior, I select (Vanilla + β -VAE) $\beta = \{1, 10, 50\}$ and (InfoVAE) $\lambda = 100$ for the DeepVAE experiments. Due to computational constraints, only a limited number of deep models are trained, with each requiring approximately one day of training on an NVIDIA 5070Ti. (need my computer for school) Architectural renderings can be seen in Figures [2, 3, 4, 5] while Section A of the appendix discloses more details regarding each block.

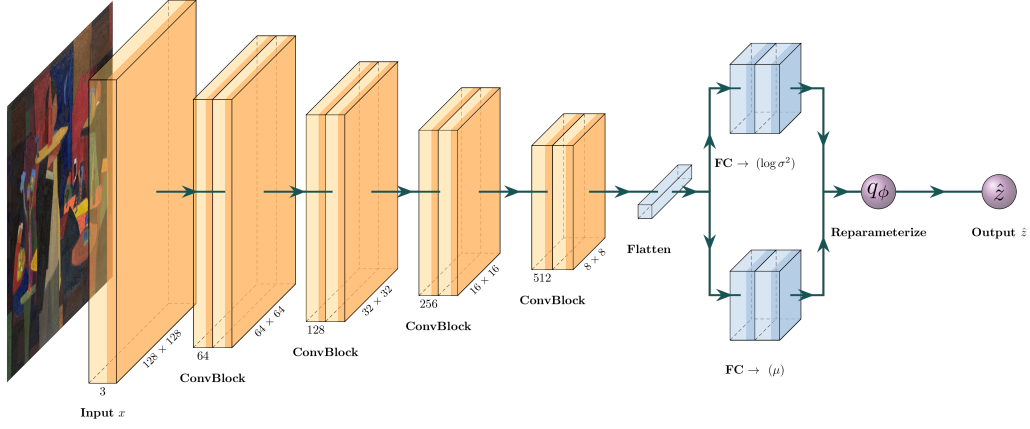


Figure 2: Architecture Diagram for VAE Encoder

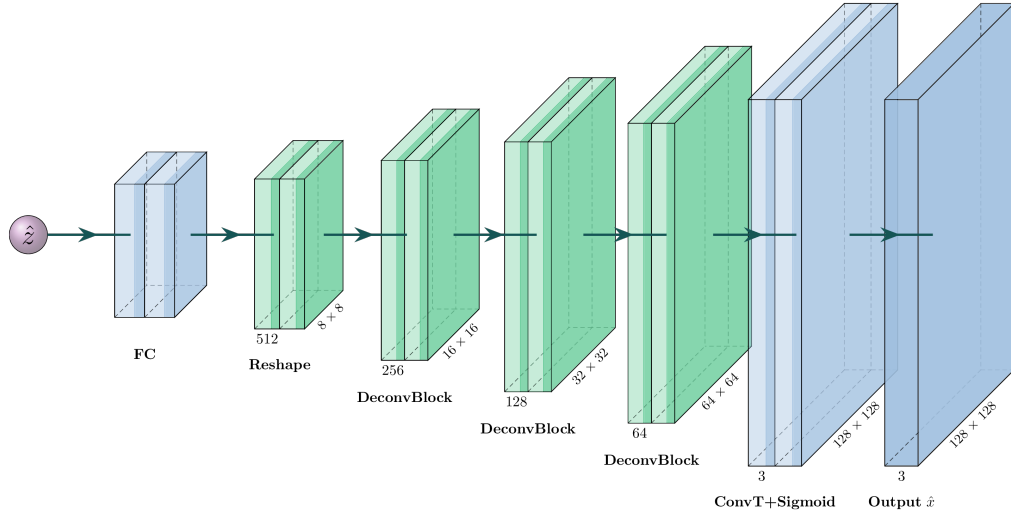


Figure 3: Architecture Diagram for VAE Decoder

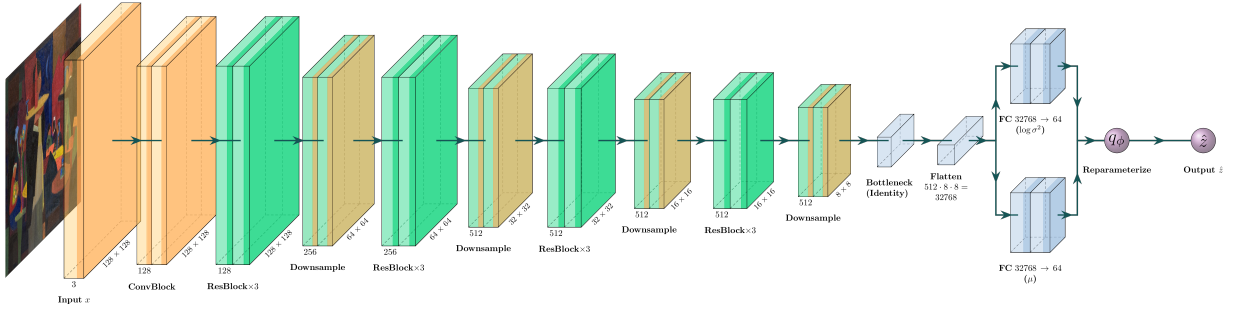


Figure 4: Architecture Diagram for DeepVAE Encoder

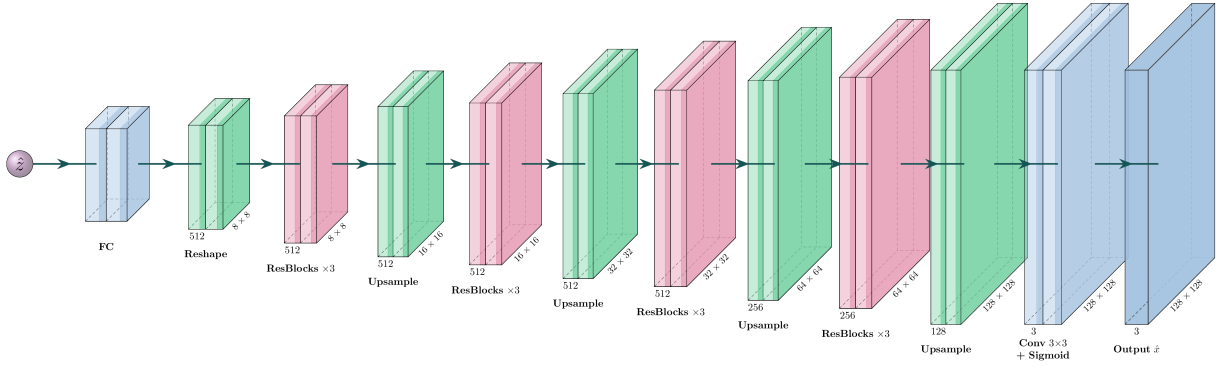


Figure 5: Architecture Diagram for DeepVAE Decoder

3.3 Training Procedure

All images were converted to RGB, resized to 128×128 using bicubic interpolation, center-cropped, and converted to tensors. Corrupted images and images larger than 80 megapixels were excluded. Exact model configurations and optimization hyperparameters are reported in Appendix Tables 6 and 7.

For unsupervised representation learning, the VAE-family models were trained on a random 90/10 train-validation split of the usable dataset under their respective generative objectives. Model selection was based on validation reconstruction loss, and the best checkpoint was retained for downstream evaluation.

Downstream classification was performed on the balanced tasks defined in Section 3.4. In all VAE-based downstream settings, classification is performed using the encoder mean μ , rather than stochastic samples from $q_\phi(z | x)$. In this sense, μ is treated as the stable latent summary of the input: a deterministic representation of the approximate posterior that is more appropriate for evaluating how well the learned bottleneck organizes class-relevant structure.

To assess representation quality, I first evaluated frozen encoders by training lightweight probe heads on extracted latent embeddings. I then considered a finetuning setting in which pretrained encoders were optimized jointly with the classifier head. Finally, I trained VAE and DeepVAE encoders directly on the classification objective to provide an end-to-end supervised comparison.

As an additional baseline, ResNet-18, ResNet-50, ResNet-101, and ResNet-152 models were trained on the same downstream tasks, both from scratch and with pretrained initialization. Across all supervised settings, model selection was based on validation Macro-F1, while final performance was reported using the metrics described in the evaluation strategy. Further implementation details, including optimizer settings, schedulers, warmup procedures, and batch sizes, are given in the appendix.

3.4 Evaluation Strategy

To evaluate performance, I construct three classification tasks of varying difficulty. The tasks differ both in the number of classes and the number of examples per class.

Task	# Classes	Samples/Class	Description
Hard	19	300	Low data, high ambiguity
Medium	15	1000	Moderate difficulty
Easy	10	2000	Higher data availability

Table 2: Classification task configurations.

Performance is measured using the Macro-F1 score, Top-1, and Top-3 accuracy. In addition to quantitative evaluation, I conduct a qualitative analysis of the learned latent spaces. This includes projecting latent embeddings into 2D using PCA, analyzing neighborhood structure via K-Nearest Neighbors, and performing

latent traversals to study how individual dimensions affect reconstruction and classification. These analyses are conducted both before and after finetuning, and compared against supervised baselines to understand how different objectives shape the organization and richness of the latent space.

4 Results

To start, we study the results of the basic classification probing performed on all trained models. I select the best performing model from each family and report their performance on each classification task. As can be seen in Table 3, the best performing models only modestly outperform the chance probability for each classification task. However, while the Top-1 and Top-3 accuracies are limited, they indicate that the latent embeddings contain *some* notion of style and can be composed into a weak classifier.

The learned representations capture simpler visual structures that are sufficient for reconstruction, but are not explicitly aligned with the higher-level abstractions required to distinguish artistic style. Despite their limited performance, these frozen-encoder models provide the clearest window into the classification process. By identifying which latent dimensions contribute most to a prediction and traversing them, we can gain an interpretable view of what the model is visually considering. We return to this idea when analyzing the latent space in more detail.

Objective	Config	Head	Architecture	Top-1	Top-3	Macro-F1	ΔAcc
Hard (19 classes)							
β -VAE	$\beta = 25$	MLP	VAE	0.221	0.389	0.225	0.186
InfoVAE	$\lambda = 250$	MLP	VAE	0.221	0.408	0.208	0.168
VAE	$\beta = 1$	MLP	VAE	0.195	0.384	0.189	0.142
Medium (15 classes)							
β -VAE	$\beta = 10$	MLP	VAE	0.259	0.511	0.251	0.192
InfoVAE	$\lambda = 75$	MLP	VAE	0.265	0.514	0.256	0.199
VAE	$\beta = 1$	MLP	VAE	0.249	0.499	0.242	0.182
Easy (10 classes)							
β -VAE	$\beta = 50$	MLP	VAE	0.291	0.553	0.277	0.191
InfoVAE	$\lambda = 100$	MLP	VAE	0.288	0.575	0.280	0.188
VAE	$\beta = 1$	MLP	VAE	0.284	0.587	0.277	0.184

Table 3: Best frozen-encoder probe per objective. Both Linear and MLP heads were considered. $\Delta\text{Acc} = \text{Acc} - 1/K$.

Following the probe experiments, I conduct a finetuning of each model. As shown in Table 4, finetuning leads to a noticeable improvement in performance, particularly on the medium and easy tasks. However, this improvement comes at the cost of interpretability, as the latent space becomes more structured for separation rather than explanation. Thus, the main advantage of pretraining via a generative task no longer affords us as much interpretability than before. Once again, we more closely inspect this tradeoff when analyzing the latent space.

Finally, Table 5 showcases the results of the baseline tests conducted on various ResNet models and the VAE and *DeepVAE* architectures. There are a few things to appreciate in these results. A first observation is that even pretrained ResNet models do not achieve near-perfect performance (i.e., $> 90\%$ Top-1 accuracy), highlighting the inherent difficulty of the task. Classifying artistic style sometimes does not correspond to clear distinctions in pixel space, making it a fundamentally challenging problem for vision models.

The classification results show that pretrained ResNet models achieve the strongest performance by a considerable margin. However, this raises a deeper question: how are these models organizing their representations, and what do they capture? By examining these differences, we can better understand what the model is visually attending to during classification. It is here that the strength of the VAE becomes apparent, as it provides a clearer window into this process.

Objective	Config	Head	Architecture	Top-1	Top-3	Macro-F1	ΔAcc
Hard (19 classes)							
β -VAE	$\beta = 50$	Linear	VAE	0.256	0.475	0.236	0.203
InfoVAE	$\lambda = 100$	Linear	VAE	0.267	0.482	0.264	0.214
VAE	$\beta = 1$	Linear	VAE	0.237	0.471	0.230	0.228
Medium (15 classes)							
β -VAE	$\beta = 10$	Linear	DeepVAE	0.349	0.636	0.342	0.282
InfoVAE	$\lambda = 75$	Linear	VAE	0.321	0.605	0.315	0.254
VAE	$\beta = 1$	Linear	DeepVAE	0.321	0.611	0.318	0.254
Easy (10 classes)							
β -VAE	$\beta = 10$	Linear	DeepVAE	0.381	0.682	0.373	0.281
InfoVAE	$\lambda = 100$	Linear	DeepVAE	0.365	0.668	0.354	0.265
VAE	$\beta = 1$	Linear	DeepVAE	0.365	0.662	0.351	0.265

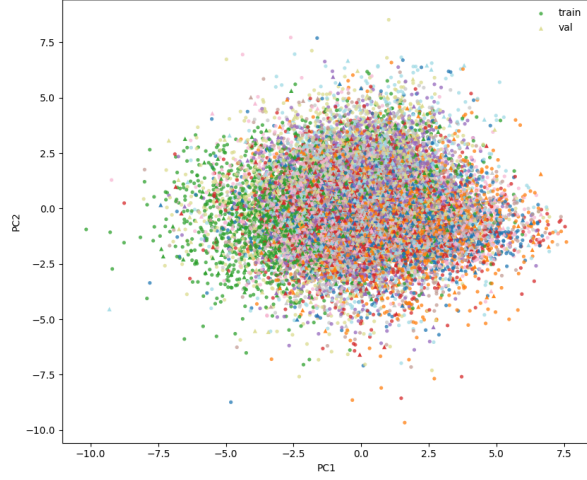
Table 4: Best finetuned models per objective. Architecture indicates whether the encoder comes from the standard VAE or DeepVAE.

Category	Model	Init	Head	Top-1	Top-3	Top-5	Macro-F1	ΔAcc
Hard (19 classes)								
ResNet	ResNet50	Pretrained	MLP	0.461	0.725	0.823	0.458	0.409
ResNet	ResNet50	Scratch	MLP	0.268	0.518	0.672	0.268	0.216
End-to-End	VAE	–	Linear	0.214	0.403	0.591	0.209	0.114
Medium (15 classes)								
ResNet	ResNet101	Pretrained	MLP	0.568	0.815	0.906	0.568	0.501
ResNet	ResNet50	Scratch	MLP	0.362	0.618	0.774	0.362	0.295
End-to-End	VAE	–	MLP	0.317	0.519	0.722	0.315	0.250
Easy (10 classes)								
ResNet	ResNet152	Pretrained	MLP	0.602	0.855	0.952	0.601	0.502
ResNet	ResNet18	Scratch	MLP	0.386	0.667	0.829	0.385	0.286
End-to-End	VAE	–	Linear	0.354	0.619	0.794	0.344	0.254

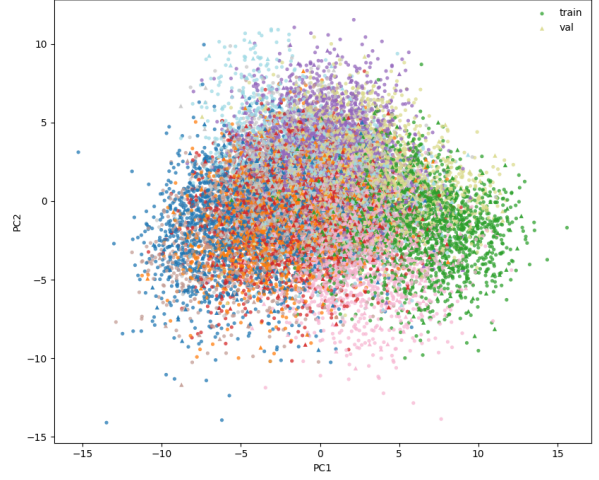
Table 5: Best baseline classifiers per task difficulty. We report the best ResNet pretrained and from-scratch models (baseline runs), and the best end-to-end classifier (e2e runs). $\Delta\text{Acc} = \text{Top-1} - 1/K$.

As a starting point, I use the vanilla *DeepVAE* model, which performed about average across most classification tasks and serves as a reasonable baseline. In Figure 6, we can visually discern the differences between non-finetuned and finetuned latent spaces after projecting all embeddings into 2D via PCA. After finetuning, the model organizes the data into more structured groups, bringing paintings of the same style closer together in latent space. As a result, dissecting and separating the data into distinct style categories becomes easier for the classification head. This is reflected in the observed $\sim 10\%$ increase in accuracy after finetuning.

This structure also provides insight into how the model differentiates between styles. For example, Baroque, Abstract Expressionism, and Impressionism (green, dark blue, and purple) form relatively distinct clusters, while Expressionist pieces (red) remain more dispersed across the latent space. This dispersion suggests that the model struggles to form a consistent representation for Expressionism, which in turn makes classification more difficult. This interpretation is supported by the confusion matrix in Figure 7 (b), where the predicted class distribution for Expressionist pieces is spread across multiple classes. This effect is even more pronounced in the non-finetuned model, as shown in Figure 7 (a), where predictions across all styles are more broadly distributed, reflecting a less structured latent space that is not well aligned with stylistic distinctions.

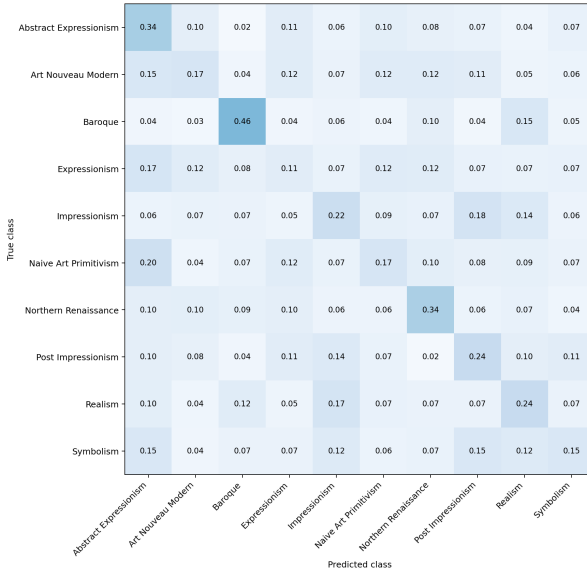


(a) Non-finetuned

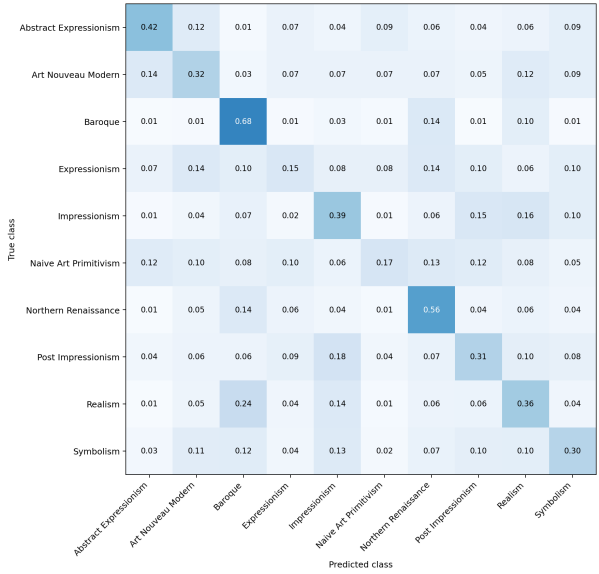


(b) Finetuned

Figure 6: Comparison of the entire dataset’s latent embeddings projected down to 2D via PCA. Each color represents a specific style of paintings where there n=10 types of styles.



(a) Non-finetuned



(b) Finetuned

Figure 7: Comparison of the Easy task dataset latent embeddings projected down to 2D via PCA. Each color represents a specific style of paintings where there K=10 types of styles.

To further reinforce this idea, we can examine Figure 8. The pretrained ResNet-152 model produces tightly clustered representations, with each style forming a well-defined group. This organization aligns with its strong classification performance ($\sim 60\%$ Top-1 accuracy), as the model learns to construct embeddings that are tailored for classification. In this sense, the latent space is shaped explicitly around the task: it compresses the data into representations that are most useful for classification prediction.

However, this structure and organization come at a cost. In forming a representation optimized for classification, multiple visual factors become entangled within a single compact encoding. Rather than isolating distinct visual elements, the model compresses them into features that are useful for predicting style but difficult to interpret individually. As a result, while performance improves, our ability to understand what

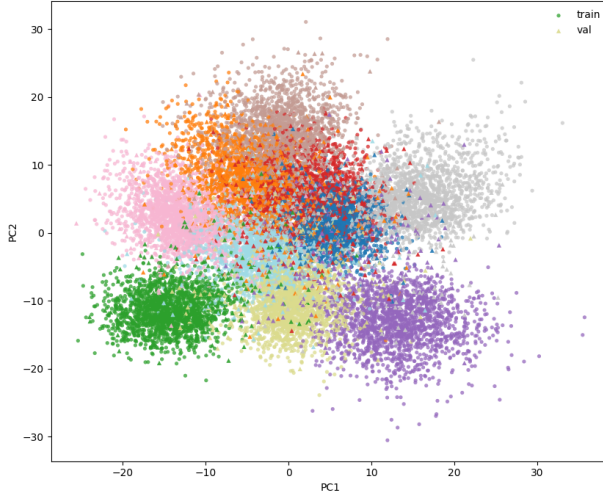


Figure 8: PCA of the latent embeddings from a pretrained ResNet152 model with $n=10$ classes

the model has learned is diminished. Unlike VAEs, these representations do not readily support exploration, as there is no clear way to traverse individual dimensions and study the abstractions they encode.

Fundamentally, what one must consider when studying these models is how much they care to know *how* the model is making predictions. A ResNet, or any other traditional supervised classification model, affords us stronger performance but limited interpretability. A VAE, or more generally, a model pretrained with a generative objective, may limit task-specific performance but improves our understanding of how a prediction is formed. In this sense, a VAE allows us to ask *why*. This naturally leads to a broader question. If a sufficiently expressive model is able to decompose a complex painting into simpler latent components, what can we learn from the forms it reduces to? Do these abstractions align with the compositional structures we intuitively recognize?

Among the models considered, the β -VAE ($\beta = 10$) *DeepVAE* is perhaps the most compelling to examine. It offers a balanced tradeoff between reconstruction quality and latent structure, producing representations that are both useful and interpretable. This is reflected in the organization of its latent space; accordingly, we evaluate classification using a frozen pretrained encoder paired with an MLP head. By traversing the latent dimensions that contribute most to classification, we begin to see the visual elements the model relies on.

Formally, the process is as follows: (1) sample a datapoint, (2) classify it, (3) identify the most influential latent dimensions in the classification, (4) traverse those dimensions, and (5) interpret the resulting variations. While the visual quality of these traversals is limited due to the scale of the model, they nonetheless provide insight into how the model is forming its representations. For each example, we present the original painting, its reconstruction, its predicted and true style, and traversals of the most influential latent dimensions. Among these, we highlight the dimensions with the lowest standard deviation in the latent encoding, as these correspond to the most stable directions in the representation. When these dimensions overlap with those contributing most to classification, they provide a clearer signal of what differentiates a painting within the latent space, and offer a more interpretable view into the abstractions the model relies on.

To ground this discussion, we highlight several examples from the latent space study. We begin with a painting from Claude Monet’s Waterloo Bridge series, specifically *Waterloo Bridge, Gray Day*. This is followed by *The Shell* by Rembrandt, *Portrait of Francesco d’Este* by Rogier Van Der Weyden, and *Landscape* by Volodymyr Orlovsky. Figures [9, 10, 11, 12] capture snapshots spanning a range of styles and visual structures.

Using Figure 12 (*Landscape*), we first identify the top-10 latent dimensions that contribute most to classification, and the top-10 dimensions with the lowest standard deviation. We then examine their overlap and traverse these shared dimensions to better understand what the model is visually attending to.



Figure 9: From left to right: Original *Waterloo Bridge, Gray Day*, its reconstruction, and the top three nearest neighbors in latent space (by cosine distance). Nearest neighbors (left to right): *Lake Hyumpola* by Nicholas Roerich (Symbolism), *During Haying* by Efim Volkov (Realism), and *The Road from Montmartre in Paris* by Ilya Repin (Realism).



Figure 10: From left to right: *The Shell (Conus Marmoreus)* by Rembrandt (Baroque), its reconstruction, and the top three nearest neighbors in latent space. Nearest neighbors (left to right): *Gear* by Anna Ostroumova-Lebedeva (Art Nouveau), *Landscape* by Kuzma Petrov-Vodkin (Symbolism), and *The Carnival: Chiarina* by Léon Bakst (Art Nouveau).



Figure 11: From left to right: *Portrait of Francesco d'Este* by Rogier van der Weyden (Northern Renaissance), its reconstruction, and the top three nearest neighbors in latent space. Nearest neighbors (left to right): *Portrait of Hieronymus Holzschuher* by Albrecht Dürer (Northern Renaissance), *Portrait of Maurice Gangnat* by Pierre-Auguste Renoir (Impressionism), and *Self-Portrait* by Kuzma Petrov-Vodkin (Symbolism).

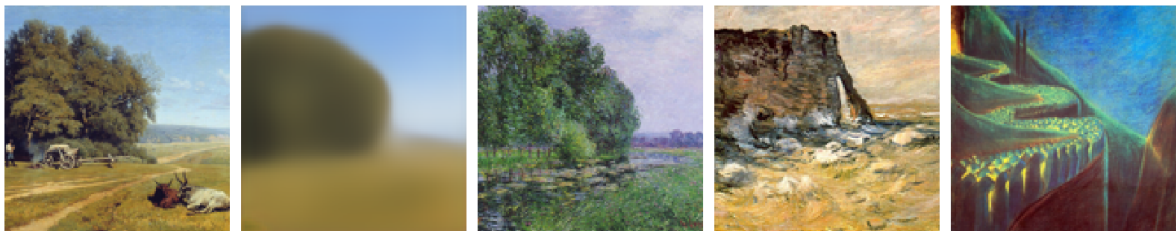


Figure 12: From left to right: *Landscape* by Volodymyr Orlovsky (Realism), its reconstruction, and the top three nearest neighbors in latent space. Nearest neighbors (left to right): *By the Eure River in Summer* by Gustave Loiseau (Post-Impressionism), *Cliffs of the Porte d'Aval* by Claude Monet (Impressionism), and *Funeral Symphony (VI)* by Mikalojus Konstantinas Čiurlionis (Symbolism).

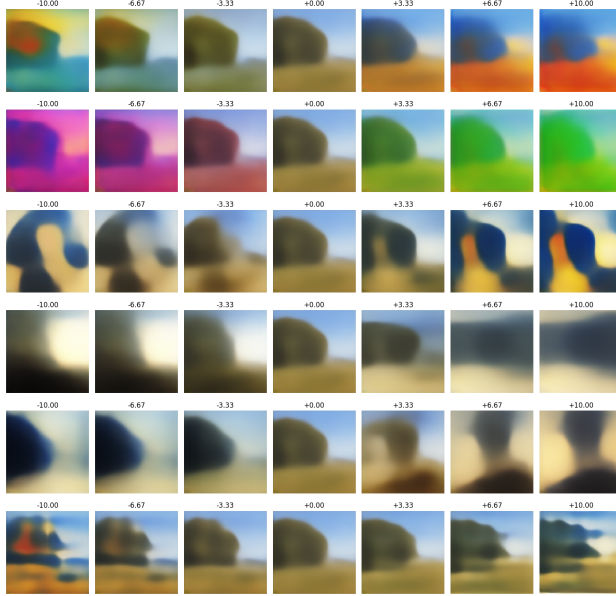


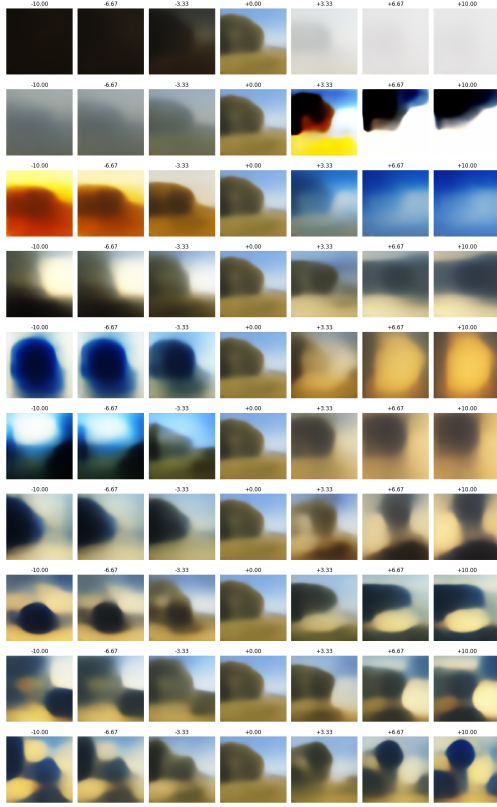
Figure 13: Architecture Diagram for VAE Convolutional Block (ConvBlock)

As shown in Figure 13, there is substantial overlap between the most discriminative and most stable dimensions. This suggests that the classifier relies on these consistent latent directions, which capture information that is both structurally meaningful and useful for prediction. Some of these dimensions also vary smoothly along simple visual properties, making them easier to interpret.

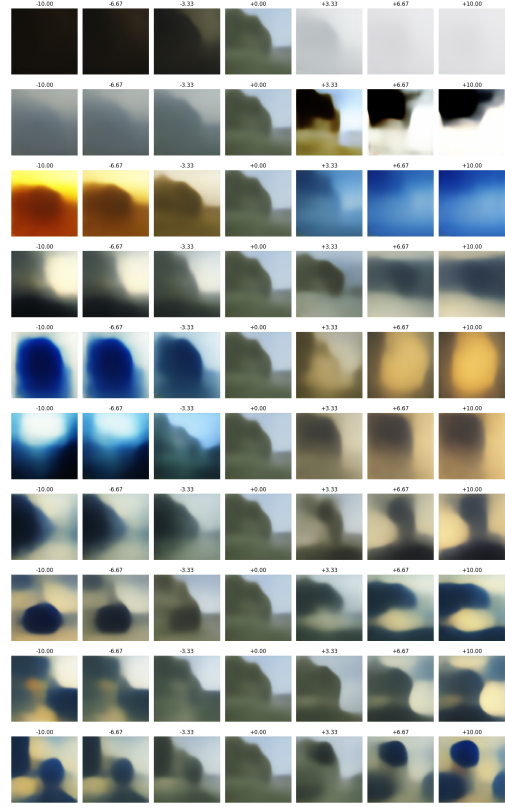
A complementary observation is that each example’s nearest neighbors exhibit clear visual similarity to the query painting. While this is encouraging, it raises a more fundamental question: why does the model consider one painting similar to another? Returning to Figure 12 and its nearest neighbor, *By the Eure River*, we find near-perfect overlap in the top-10 lowest standard deviation dimensions, with only minor differences in ranking. As shown in Figure 14 (a) and (b), by traversing these axes we can identify that some correspond to smooth changes in color, lighting, and aspects of spatial structure. In this sense, similarity is grounded in shared variations of basic visual properties.

We also examine the top-10 lowest absolute error latent dimensions between the query and its nearest neighbor. As shown in Figure 14 (c), some of these dimensions exhibit simple and interpretable patterns, while others are more difficult to attribute to a single visual factor. This highlights a second notion of similarity: one based on proximity in latent space rather than alignment along specific directions. Taken together, these results suggest that similarity arises from two sources: alignment along coherent, interpretable latent directions, and closeness under the latent distance. The former provides a clearer explanation of shared visual structure, while the latter reflects how the model internally measures similarity, even when it is less easily interpretable.

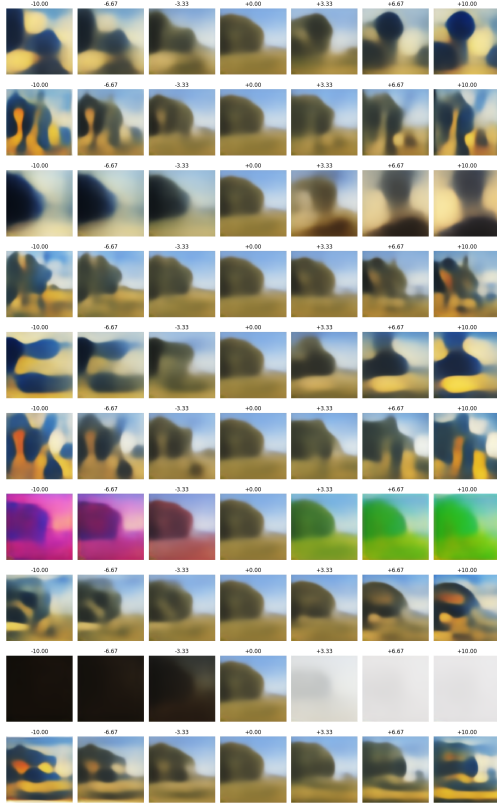
Finally, we examine whether these latent directions are consistent across different styles. By traversing the same low-variance dimensions identified from the landscape painting on a portrait example, we observe that their behavior remains largely consistent. As shown in Figure 14 (d), divergence appears primarily in higher-variance dimensions. This suggests that low-variance dimensions capture general, style-agnostic properties such as global color and forms, while higher-variance dimensions encode more specialized structure. In this sense, the latent space appears to be organized along a spectrum of abstraction, from simple and general visual factors to more complex and style-specific representations.



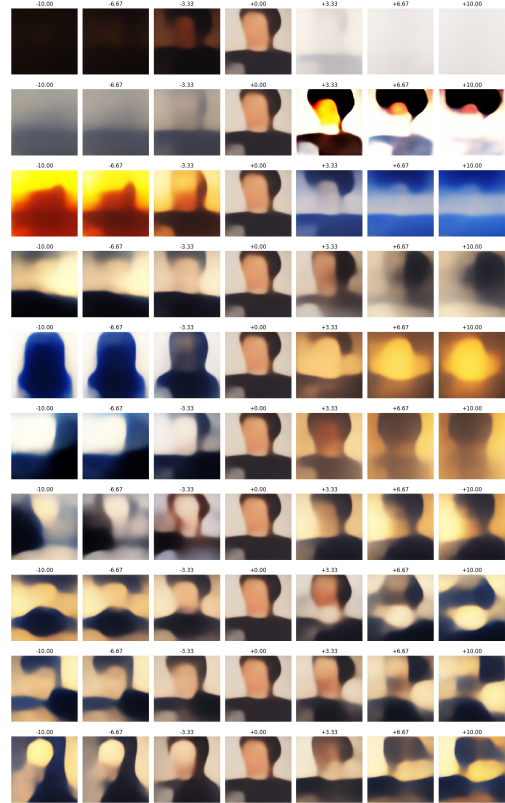
(a) Landscape traversal (low-var dims)



(b) NN traversal (same dims)



(c) NN traversal (low-error dims)



(d) Portrait traversal (same dims)

Figure 14: Latent traversals across shared and low-error dimensions.

It is worth noting that this analysis remains primarily qualitative. While many of the observed patterns in the latent space suggest directions for more formal, quantitative study, I deliberately focus here on building intuition through visualization and simple probing. Establishing this connection between latent structure, classification behavior, and traversal-based interpretation provides a foundation for deeper analysis, while leaving several promising directions for future work.

5 Conclusion

In this work, we examined how different learning objectives shape the structure and usefulness of latent representations. While supervised models such as ResNets produce highly organized latent spaces that are well-suited for classification, this organization comes at the cost of interpretability. In contrast, VAEs, particularly the β -VAE ($\beta = 10$), offer a more interpretable latent space that can be directly explored through traversal. Through quantitative and qualitative analyses, we find that VAE representations capture meaningful visual structure, even if they do not achieve near perfect classification performance.

More importantly, this study highlights a fundamental tradeoff between performance and interpretability. Models trained purely for prediction learn representations that are efficient but opaque, whereas models trained under a generative objective expose a structure that can be interrogated and understood. By leveraging latent traversals, we demonstrate that similarity in latent space arises not only from proximity, but also from shared directions of variation that correspond to basic visual properties.

Ultimately, this project does not aim to propose a new method, but rather to provide a lens through which representation learning can be studied. VAEs offer a particularly appealing framework in domains such as art, where structure is rich, compositional, and inherently open to interpretation. In this setting, learned representations become more than tools for prediction; they offer a way to explore how visual abstractions are formed and related.

A Architecture Figures

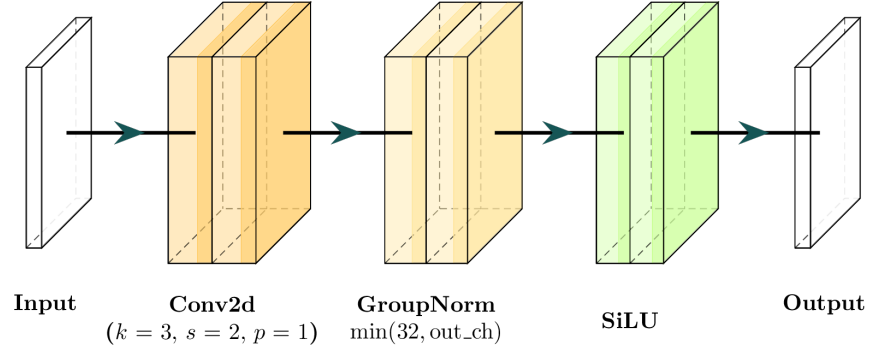


Figure 15: Architecture Diagram for VAE Convolutional Block (ConvBlock)

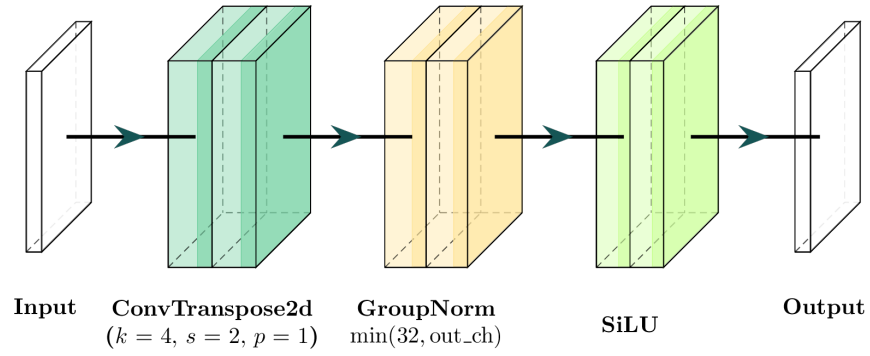


Figure 16: Architecture Diagram for VAE Deconvolutional Block (DeconvBlock)

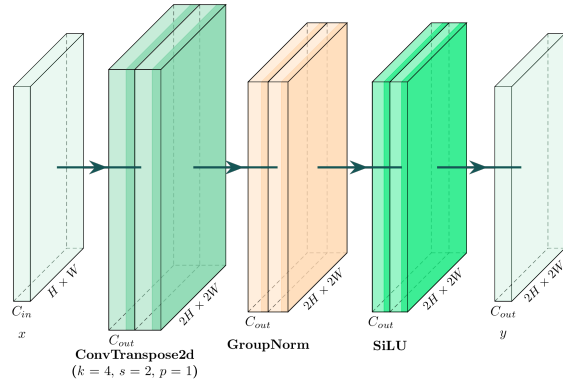


Figure 17: Architecture Diagram for DeepVAE Upsample Block (Upsample)

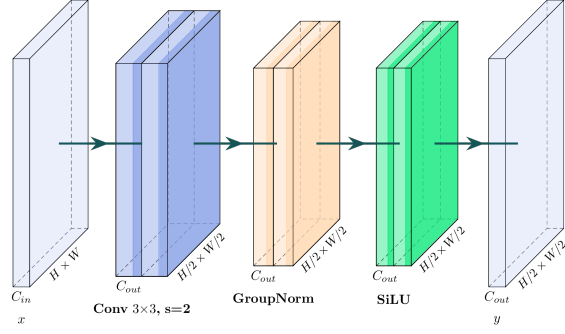


Figure 18: Architecture Diagram for VAE Downsample Block (Downsample)

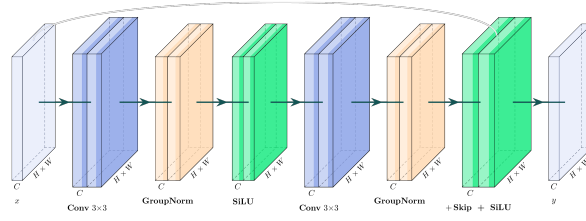


Figure 19: Architecture Diagram for VAE Residual Block (ResBlock)

B Other Latent Space Figures

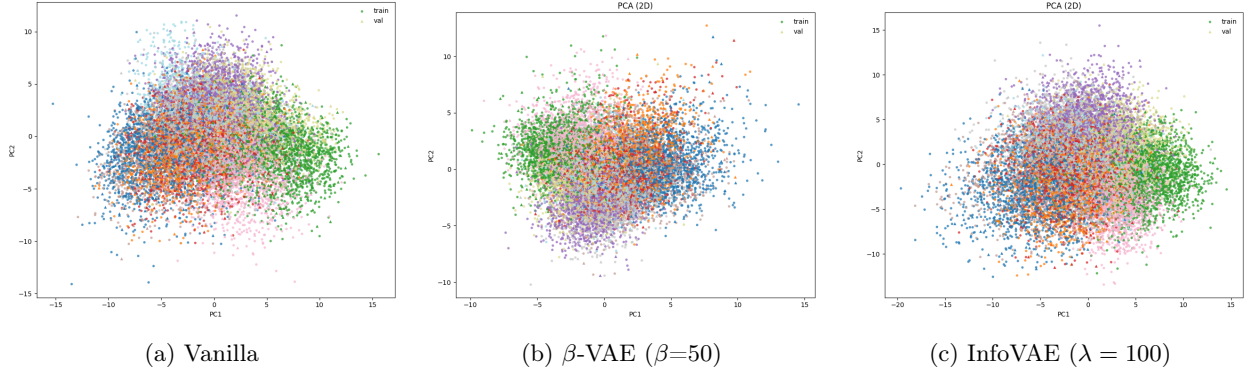
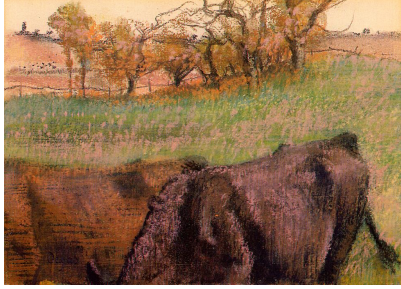


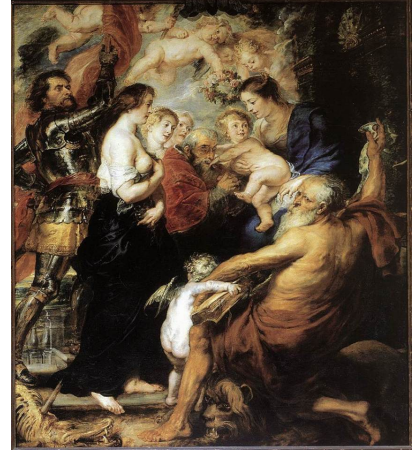
Figure 20: Comparison of Finetuned Latent Embeddings for Different VAE Model Families



(a) *Edgar Degas - Landscape. Cows in the Foreground* (Impressionism)

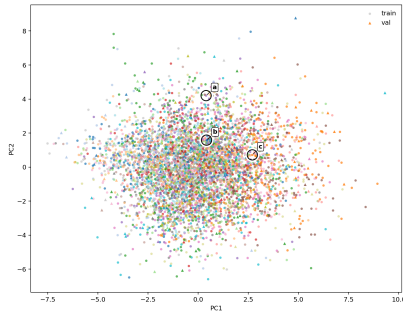


(b) *Sam Francis - Untitled (sfp88)* (Abstract Expressionism)

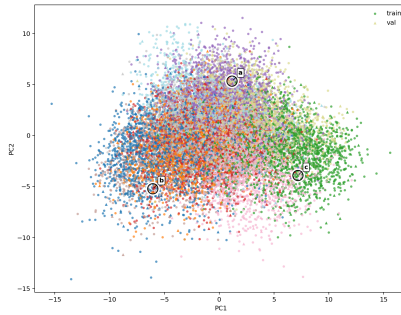


(c) *Peter Paul Rubens - Our Lady with the Saints* (Baroque)

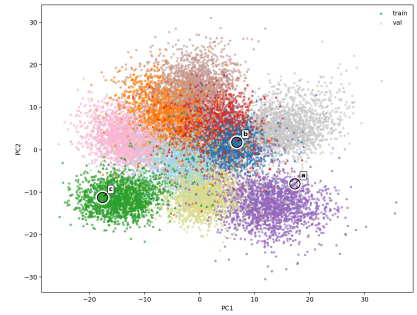
Figure 21: Selected Paintings Used to Compare Latent Space Placement Across Models



(a) Vanilla *DeepVAE*



(b) Finetuned Vanilla *DeepVAE*



(c) Pretrained ResNet 152

Figure 22: Comparison of Latent Space Placement Across Different Models

C Training Hyperparameters

Component	Setting	Value
Shared VAE-family pretraining settings		
Pretraining	Preprocessing	RGB conversion, bicubic resize to 128×128 , center crop, tensor conversion
Pretraining	Data split	Random 90/10 train-validation split
Pretraining	Optimizer	AdamW
Pretraining	Learning rate / weight decay	2×10^{-4} / 0
Pretraining	Epochs	100
Pretraining	Reconstruction loss	L_1
Pretraining	KL warmup	2000 steps for VAE and β -VAE
Pretraining	Gradient clipping	1.0
Pretraining	Checkpoint selection	Lowest validation loss
Pretraining	Seed	0
VAE-family sweep definitions		
Vanilla VAE	Backbone	VAE, $z = 64$, base channels = 64, max channels = 512, 4 layers
Vanilla VAE	Objective / batch	$\beta = 1$, batch size 64, AMP off
β -VAE sweep	Backbone	Same as vanilla VAE
β -VAE sweep	Objective / batch	$\beta \in \{5, 10, 25, 50\}$, batch size 128, AMP off
InfoVAE sweep	Backbone	Same as vanilla VAE
InfoVAE sweep	Objective / batch	$\alpha = 0$, $\lambda \in \{50, 75, 100, 250, 500\}$, MMD sigmas $\{1, 2, 4, 8, 16\}$, batch size 64 to 1024, AMP on for reported runs
DeepVAE variants	Backbone	DeepVAE, $z = 64$, base channels = 128, max channels = 512, 4 stages, 3 blocks/stage
DeepVAE variants	Batch / AMP	Batch size 24, AMP on

Table 6: Pretraining hyperparameters for the VAE-family models. Shared settings are listed once; objective- and architecture-specific sweeps are summarized separately.

Protocol	Model / head	Hyperparameters
Frozen probe	Fixed encoder mean μ ; linear or 1-hidden-layer MLP	AdamW, lr = 10^{-3} , wd = 0, batch = 128, hidden dim = 256, dropout = 0.1, 200 epochs max, patience = 50, best by val Macro-F1
Finetuning	Pretrained encoder + linear or 1-hidden-layer MLP	2 warmup epochs with frozen encoder, then joint training; AdamW, head lr = 10^{-3} , encoder lr = 10^{-4} , wd = 10^{-4} , batch = 64^3 , hidden dim = 256, dropout = 0.1, ReduceLROnPlateau, patience = 8, best by val Macro-F1
End-to-end VAE classifier	VAE or DeepVAE encoder trained directly for classification; linear or 1-hidden-layer MLP	AdamW, lr = 10^{-3} , wd = 10^{-4} , batch size architecture-dependent, hidden dim = 256, dropout = 0.1, 120 epochs max, ReduceLROnPlateau, patience = 20, best by val Macro-F1
ResNet baselines	ResNet-18/50/101/152, scratch and pretrained; 64-d bottleneck + 2-layer MLP	AdamW, lr = 10^{-3} , wd = 10^{-4} , batch = 128, latent dim = 64, dropout = 0.1, 100 epochs, ReduceLROnPlateau, patience = 15, best by val Macro-F1

Table 7: Downstream classification hyperparameters. All downstream experiments use the balanced evaluation tasks defined in the evaluation strategy section.

³For finetuning, the effective batch size is capped by the original pretraining batch size for some deeper checkpoints.

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